

MILK RIVER RIDGE RESERVOIR, AB, Canada

WMO: 710150

Lat: 49.2839N

Lon: 112.5383W

Elev: 4281

StdP: 12.56

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.3	-12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	84.3	N/A	80.8	N/A	77.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-22.3	89.8	6.8	4.3	-27.1	92.9	-31.1	95.4	-34.9	97.9	-39.8	101.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	22.5	18.7	28.6	38.4	49.0	55.8	63.5	62.7	54.1	41.6	31.7	20.0		
	DBStd	19.09	14.32	15.73	12.80	9.80	8.47	6.58	6.33	6.95	9.71	10.04	10.78	14.15		
	HDD50	4742	853	878	665	362	125	20	1	3	64	290	552	929		
	HDD65	9012	1318	1297	1130	797	497	282	103	126	345	724	999	1394		
	CDD50	1342	0	0	0	15	94	195	420	396	188	30	4	0		
	CDD65	137	0	0	0	0	1	7	57	54	18	0	0	0		
	CDH74	1558	0	0	0	3	26	68	625	622	210	4	0	0		
	CDH80	379	0	0	0	0	0	8	163	165	43	0	0	0		
	CDH80	379	0	0	0	0	0	8	163	165	43	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	14.9	0.5	0.4	0.7	1.2	2.2	3.5	1.6	1.4	1.3	0.9	0.6	0.6		
	PrecMax	22.8	0.8	0.9	1.6	2.4	4.2	8.9	5.5	3.3	3.7	2.7	1.3	1.1		
	PrecMin	8.0	0.1	0.0	0.2	0.5	0.6	0.6	0.1	0.0	0.1	0.2	0.2	0.0		
	PrecStd	3.5	0.2	0.2	0.4	0.5	1.0	2.0	1.2	0.8	0.9	0.6	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	50.4	56.4	71.5	77.2	80.8	89.3	88.5	84.8	71.9	61.0	49.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	45.3	44.7	51.1	63.1	73.1	75.3	84.4	84.4	80.1	64.7	53.0	43.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	41.9	41.0	47.4	57.7	68.8	72.4	80.6	80.9	75.2	59.7	49.1	39.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	39.3	38.1	43.0	53.0	64.2	68.7	77.0	77.1	69.9	55.1	45.3	36.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.4	15.9	18.5	18.9	18.8	21.8	22.6	20.5	17.3	15.3	13.7		
		MCDBR	15.3	17.3	20.5	25.4	25.0	23.0	26.1	28.1	27.6	23.2	19.7	15.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.272	0.305	0.317	0.316	0.318	0.334	0.300	0.270	0.247	0.227		
	taud		2.470	2.500	2.503	2.419	2.436	2.486	2.489	2.414	2.488	2.552	2.517	2.465		
	Ebn at Noon		271	295	303	300	299	299	297	286	288	282	266	262		
	Edh at Noon		18	23	27	33	34	33	32	33	28	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		394	712	1103	1524	1815	2029	2254	1799	1278	792	441	315		
	RadStd		32	53	96	100	158	155	123	113	105	77	34	30		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page